Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which of the following is not a condition for complex formation?

(A) Small atomic size

(B) High nuclear charge

(C) Variable oxidation states

(D) Availability of vacant d orbitals

Q.2

Zn gives H_2 gas with H_2SO_4 and HCl but not with HNO_3, because

(A) Zn acts as an oxidising agent when react with HNO_3.

(B) HNO_3 is weaker acid than H_2SO_4 and HCl.

(C) In electrochemical series Zn is above hydrogen.

(D) NO_3^- ion is reduced in preference to hydronium ion.

Q.3

Which of the following is not correct about transition metals?

(A) Their melting and boiling points are high

(B) Their compounds are generally coloured

(C) They can form ionic or covalent compounds

(D) They do not exhibit variable valency

Q.4

Zinc and mercury do not show variable valency like d -block element because

(A) they are soft.

(B) their d -shells are complete.

(C) they have only two electrons in the outermost shell.

(D) their d -shells are incomplete.

@IITJEE_Advanced

Q.5

Which one of the following is diamagnetic ion?

(A) Co^{2+}

(B) Cu^{2+}

(C) Mn^{2+}

(D) Sc^{3+}

Q.6

Which group contains coloured ions out of

1. Cu^{2+} 2. Ti^{4+}
3. Co^{2+} 4. Fe^{2+}

(A) $1, 2, 3, 4$

(B) $1, 3, 4$

(C) 2,3

(D) 1,2

Q.7

A compound of a metal ion $M^{x+}(Z = 24)$ has a spin only magnetic moment of $\sqrt{15}$ Bohr Magnetons. The number of unpaired electrons in the compound are

(A) 2

(B) 4

(C) 5

(D) 3

Q.8

Cuprous ion is colourless while cupric ion is coloured because

(A) both have half filled p -and d -orbitals.

(B) cuprous ion has incomplete d -orbital and cupric ion has a complete d -orbital.

(C) both have unpaired electrons in the d -orbitals.

(D) cuprous ion has complete d -orbital and cupric ion has an incomplete d -orbital.

Q.9

Among the following, the compound that is both paramagnetic and coloured, is

@IITJEE_Advanced

(A) $KMnO_4$

(B) CuF_2

(C) $K_2Cr_2O_7$

(D) All are coloured

Q.10

Mercury is the only metal which is liquid at $0^\circ C$. This is due to its

(A) very high ionisation energy and weak metallic bond.

(B) low ionisation potential.

(C) high atomic weight.

(D) high vapour pressure.

Q.11

The catalyst used in the Haber process for the manufacture of ammonia is

(A) Pt

(B) Pd

(C) Fe

(D) $\mathbf{Ni}$

Q.12

Which of the following arrangements does not represent the correct order of the property stated against it?

(A) $V^{2+} < Cr^{2+} < Mn^{2+} < Fe^{2+}$: Paramagnetic behavior

(B) $Ni^{2+} < Co^{2+} < Fe^{2+} < Mn^{2+}$: Ionic size

(C) $Co^{3+} < Fe^{3+} < Cr^{3+} < Sc^{3+}$: Stability in aqueous solution

(D) $Sc < Ti < Cr < Mn$: Number of oxidation states

Q.13

The Irving-Williams order regarding the stability of the complexes of bivalent metal of the firg: transition series is

(A) $Mn < Fe < Co < Ni < Cu > Zn$

(B) $Mn > Fe > Co > Ni > Cu > Zn$

(C) $Mn < Fe < Co > Ni > Cu > Zn$

(D) $Mn > Fe > Co < N < Cu < Zn$

@IITJEE_Advanced

Q.14

Which of the following ions is expected to be diamagnetic ?

(A) $25\mathrm{Mn}^{2+}$

(B) $29\mathrm{Cu}^{+}$

(C) $27\mathrm{Co}^{3+}$

(D) $29\mathrm{Cr}^{3+}$

Q.15

Which of the following statements about the complex $\left[\mathrm{CoF}_6\right]^{3-}$ which is paramagnetic in nature is correct?

(A) Cobalt involves $\mathrm{d}^2\mathrm{sp}^3$ hybridization

(B) Cobalt involves sp^3d^2 hybridization

(C) $\mathbf{F}$ is in zero oxidation state

(D) Co is in $+2$ oxidation state

Q.16

A transition element containing only one electron in 4s-orbital is:

(A) ${}_{21}\mathrm{Se}$ write this as questio

(B) ${}_{23}\mathrm{Mn}$

(C) ${}_{26}\mathrm{Fe}$

(D) ${}_{29}\mathrm{Cu}$

Q.17

The basic nature of the monoxides of the first transition series is of the following order

(A) $\mathrm{CrO} > \mathrm{VO} > \mathrm{FeO} > \mathrm{TiO}$

(B) $\mathrm{T_OO} > \mathrm{VO} > \mathrm{CrO} > \mathrm{FeO}$

(C) $\mathrm{VO} > \mathrm{Cr}O > \mathrm{TiO} > \mathrm{FeO}$

(D) $\mathrm{TO} >> \mathrm{FeO} > \mathrm{CrO} > \mathrm{VO}$

Q.18

Maximum magnetic moment is shown by:

(A) $\mathbf{d}^5$

(B) d^6

@IITJEE_Advanced

(C) d^7

(D) d^8

Q.19

Zinc reacts with excess of caustic soda to form

(A) $Zn\,(OH_2$

(B) ZnO

(C) Na_2ZnO_2

(D) $Zn(OH)2ZnCO_3$

Q.20

Uranium can be obtained by the electrolysis of:

(A) UF_6

(B) UF_4

(C) UCl_6

(D) UI_6

Answers & Solutions

Q.1 Answer:

Variable oxidation states

Solution:

(c) Tendency of an element to form complexes does not depend on whether the element shows variable O.S. or not

Q.2 Answer:

NO_3^- ion is reduced in preference to hydronium ion.

Solution:

(d) NO_3^- **ions are reduced by nascent hydrogen Metal** $+HNO_3 \rightarrow$ **Metal nitrate** $+[H]$

$HNO_3 + 8[H] \rightarrow N_2O + 5H_2O$

Q.3 Answer:

They do not exhibit variable valency

Solution:

(d) Transition metals exhibit variable valency.

@IITJEE_Advanced

Q.4 Answer:

their d -shells are complete.

Solution:

(b) Zn and Hg have d -shells completely filled hence they do not exhibit variable valency.

Q.5 Answer:

Sc^{3+}

Solution:

(d) Sc^{3+} is diamagnetic as it does not contain any unpaired electron while others contain.

Q.6 Answer:

$1, 3, 4$

Solution:

(b) $\mathrm{Cu}^{2+}[\mathrm{Ar}]3d^{9}, \mathrm{Ti}^{4+}[\mathrm{Ar}]3d^{0}, \mathrm{Co}^{2+}[\mathrm{Ar}]3d^{7}$
$\mathrm{Fe}^{2+}[\mathrm{Ar}]3d^{6} 1, 3, 4$ are coloured ions
hence the answer is b.

Q.7 Answer:

3

Solution:

(d) Magnetic moment $\mu = \sqrt{\mathrm{n}(\mathrm{n}+2)}$ where n = number of unpaired electrons $\sqrt{15} = \sqrt{\mathrm{n}(\mathrm{n}+2)} \therefore \mathrm{n} = 3$

Q.8 Answer:

cuprous ion has complete d -orbital and cupric ion has an incomplete d -orbital.

Solution:

(d) In $\mathrm{Cu}^{+}[\mathrm{Ar}]3\ \mathrm{d}^{10}$ there is no unpaired electron, $\mathrm{Cu}^{2+}[\mathrm{Ar}]3\ \mathrm{d}^{9}$ contains one unpaired electron hence coloured.

Q.9 Answer:

CuF_2

Solution:

CuF_2 is both paramagnetic and coloured.

Q.10 Answer:

very high ionisation energy and weak metallic bond.

Solution:

@IITJEE_Advanced

Metallic bonding in mercury is weak.

Q.11 Answer:

Fe

Solution:

Fe is used as a catalyst in Haber process.

Q.12 Answer:

$\mathrm{V}^{2+} < \mathrm{Cr}^{2+} < \mathrm{Mn}^{2+} < \mathrm{Fe}^{2+}$: Paramagnetic behavior

Solution:

Greater is the number of unpaired electrons, greater is the paramagnetic behavior.

$\mathrm{V} = 3\,\mathrm{d}^3 4\,\mathrm{s}^2; \mathrm{V}^{2+} = 3\,\mathrm{d}^3 = 3$ **unpaired electrons**

$\mathrm{Cr} = 3\,\mathrm{d}^5 4\,\mathrm{s}^1; \mathrm{Cr}^{2+} = 3\,\mathrm{d}^4 = 4$ **unpaired electrons**

$\mathrm{Mn} = 3\,\mathrm{d}^5 4\,\mathrm{s}^2$

$\mathrm{Mn}^{2+} = 3\,\mathrm{d}^5 = 5$ **unpaired electrons**

$\mathrm{Fe} = 3\,\mathrm{d}^6 4\,\mathrm{s}^2; \mathrm{Fe}^{2+} = 3\,\mathrm{d}^6 = 4$ **unpaired electrons**

Hence the correct order of paramagnetic behavior. $\mathrm{V}^{2+} < \mathrm{Cr}^{2+} \simeq \mathrm{Fe}^{2+} < \mathrm{Mn}^{2+}$

For the same oxidation state, the ionic radii generally decreases as the atomic number increases in a particular transition series, hence the order is $\mathrm{Mn}^{2+} > \mathrm{Fe}^{2+} > \mathrm{Co}^{2+} > \mathrm{Ni}^{2+}$

In solution, the stability of the compound depends upon electrode potentials, SEP of the transition metal ions are given as

$\mathrm{Co}^{3+}/\mathrm{Co} = +1.97,\ \mathrm{Fe}^{3+} = +0.77$

$\mathrm{Cr}^{3+}/\mathrm{Cr}^{2+} = -0.41, \mathrm{Sc}^{3+}$ **is highly stable as it does not show** $+2\mathrm{O.S}$

More positive is the SEP, less stable is the ion in the aqueous solution. The increasing order of stability in aqueous solution is $\mathrm{Co}^{3+} < \mathrm{Fe}^{3+} < \mathrm{Cr}^{3+} < \mathrm{Sc}^{3+}$:

$\mathrm{Sc} - (+2)(+3)$ **has two oxidation states.**

$\mathrm{Ti} - (+2), (+3), (+4)$ **has three oxidation states.**

$\mathrm{Cr} - (+2), (+3), (+4), (+5), (+6)$ **has five oxidation states.**

$\mathrm{Mn} - (+2), (+3), (+4), (+5), (+6), (+7)$ **has six oxidation states.**

The increasing order for the number of oxidation states is $\mathrm{Sc} < \mathrm{Ti} < \mathrm{Cr} < \mathrm{Mn}$.

Hence, option A is correct.

Q.13 Answer:

$\mathrm{Mn} < \mathrm{Fe} < \mathrm{Co} < \mathrm{Ni} < \mathrm{Cu} > \mathrm{Zn}$

Solution:

For a given ligand the stability of the complexes of the metallic ions having the same chaxge decreases with the increase in the size of the central metal ion. The sequence of the given ions in is $\mathrm{Mn}^{2+} < \mathrm{Fe}^{2+} < \mathrm{Co}^{2+} < \mathrm{Ni}^{2+} < \mathrm{Cu}^{2+} > \mathrm{Zn}^{2+}$**. This sequence is known as Irving-Williem**

@IITJEE_Advanced

order of
stability of the complexes of M^{2+} ions.

Q.14 Answer:

$29Cu^{+}$

Solution:

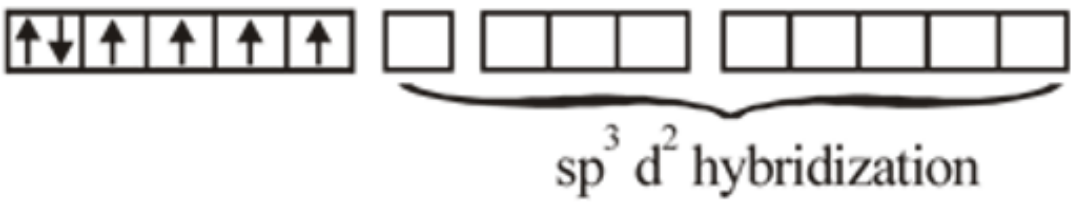
The electronic configurations are:
$_{25}Mn^{2+}(3\ d)^{5}; {}_{29}Cu^{+}(3\ d)^{10}; {}_{27}Co^{3+}(3\ d)^{6}; {}_{29}Cr^{3+}(3\ d)^{3}$. $_{29}Cu^{+}$ **has no unpaired electrons, so it is diamagnetic**

Q.15 Answer:

Cobalt involves sp^3d^2 hybridization

Solution:

Cobalt in the complex exists as



Q.16 Answer:

$_{29}Cu$

Solution:

$_{29}Cu$ **has the electronic structure of** $1\ s^2 2\ s^2 2p^6 3\ s^2 3p^6 4\ s^1 3\ d^{10}$. **One electron moves from the 45 -orbital to attain a more stable state with completely filled** d **orbital.**

Q.17 Answer:

$T_OO > VO > CrO > FeO$

Solution:

Basic strength decreases across the period.

Q.18 Answer:

d^5

Solution:

d^5 **contains 5 unpaired electrons, hence it shows maximum mapnetic moment**

Q.19 Answer:

@IITJEE_Advanced

Na_2ZnO_2

Solution:

$Zn + NaOH \longrightarrow Zn\,(OH_2$

$Zn\,(OH_2 + NaOH \longrightarrow Na_2\,[Zn(OH)_4]$

or $Na_2ZnO_2 \cdot 2H_2O$

Q.20 Answer:

UF_4

Solution:

..

@IITJEE_Advanced